

SOP-SK-04 — Technology Scouting

Kode	SOP-SK-04
Pemilik	Senku (R&D Engineer)
Revisi	1.0 · Berlaku 2026-05-29 · Reviu per kuartal
Referensi	Tech radar (Adopt/Trial/Assess/Hold), tools/6q-critical-filter.md
COBIT	APO04 (Managed Innovation)

1. Tujuan

Mengevaluasi kandidat teknologi baru (library, vendor, pattern, integrasi) secara objektif lewat **6Q critical filter**, supaya adopsi berbasis substansi & evidence — **bukan hype**. Hasilnya: **TADR** (Tech Adoption Decision Record).

2. Ruang Lingkup

- **Berlaku**: tiap kandidat tech yang dipertimbangkan masuk stack/produk (dependency baru, vendor, pattern arsitektur, integrasi pihak ketiga).
- **Tidak berlaku**: utility kecil yang reversible & lokal tanpa lock-in (cukup judgement dev di code review).

3. Definisi & Istilah

- **Tech radar** — status teknologi: **Adopt** (pakai), **Trial** (coba terbatas), **Assess** (riset dulu), **Hold** (jangan/usang).
- **Community health** — sinyal sustainability: stars, last commit cadence, jumlah contributor, release rhythm.
- **Vendor lock-in** — ketergantungan yang mahal/sulit di-undo.
- **TADR** — catatan keputusan adopsi/refute, format mirip ADR + governance section.

4. Referensi

Tata mandate (critical thinking continuous, anti adopt hype), Kakashi reversibility framework (Type-1/Type-2), evidence hierarchy.

5. Tanggung Jawab (RACI)

Aktivitas	R	A	C	I
Evaluasi & TADR	Senku	Senku	@kakashi (feasibility/reversibility), @saitama (BE fit)	tim
Keputusan adopsi (produk)	Senku (recommend)	Tata	@kakashi, @lelouch	tim
Type-1 lock	—	Tata	@kakashi (bruce+archrival)	tim

6. Prosedur

6.1 Intake

- 6.1.1 Catat kandidat: nama, versi, **license** (MIT/Apache/GPL/Commercial), link.
- 6.1.2 Audit **community health** — stars, last commit, contributor count, release cadence. Repo mati = sinyal merah.
- 6.1.3 Set status awal tech radar: **Assess**.

6.2 6Q critical filter

- 6.2.1 Lewatkan kandidat via **6 pertanyaan** (6q-critical-filter): 1. Solve real problem? (trace ke user pain — bukan "cool tech") 2. Sustainable? (maintenance, lock-in, community, longevity) 3. Fit stack? (React 18, Chakra v2, Fauxbase, migration cost) 4. **Compliant?** (UU PDP/GDPR/PCI/ISO — kalau sentuh data/payment → trigger SOP-SK-03) 5. Business-fit? (ROI, unit economics, customer impact) 6. **Reversible?** (Type-1 → escalate + bruce/archrival; Type-2 → cepat)
- 6.2.2 **Exit kalau Q1 = ❌** (solution-looking-for-problem) → **refute**, stop. Gak usah lanjut filter.

6.3 Keputusan & TADR

- 6.3.1 **Build vs Buy vs Wait**: build kalau core + no vendor mature + < 1 dev-month; buy kalau commodity + vendor mature + > 3 dev-month; else wait.
- 6.3.2 Tulis **TADR** (Tech Adoption Decision Record) di `adr/NNN-*.md` — context, candidate, 6Q filter table, governance/compliance assessment, alternatives, recommendation, constraint.
- 6.3.3 Update **tech radar** ring (Adopt/Trial/Assess/Hold).
- 6.3.4 **Decision point**: Type-1 (lock vendor/pattern/data-shape)? → **escalate** 🟡 **ke Tata** via @kakashi (bruce + archrival), tunggu sign-off. Type-2 → rekomendasi 🟡.
- 6.3.5 **Exit criteria**: 6Q filter lengkap + governance/compliance assessed + verdict (adopt/constraint/refute/defer) + TADR tersimpan + radar ter-update.

7. Pengecualian

- **Tech udah dipakai di project lain (proven internal)**: boleh fast-track, **tapi tetap cek compliance** kalau context data beda.

8. Rekaman & Retensi

Rekaman	Lokasi	Retensi
TADR	adr/NNN-*.md	Permanen
Tech radar	log.md / living doc	Living
6Q filter verdict	TADR	Permanen

9. Indikator Kinerja (KPI)

KPI	Rumus	Target
Hype-adopt rate	# tech diadopsi tanpa 6Q filter	0
TADR coverage	# adopsi ber-TADR ÷ total adopsi	100%
Adopsi yang di-revert < 3 bulan	# tech ditarik balik cepat	minimal

10. Riwayat Revisi

Rev	Tanggal	Perubahan
1.0	2026-05-29	Terbit pertama. Contoh acuan: AI feature → defer post-PMF (premature, Type-1 positioning)